



Topic 20

Induced Voltages and Inductance

Reading Question 20-1 (1 of 2)

A loop is placed next to a circuit containing a solenoid and a switch. The circuit is attached to an AC source and then the switch is closed. Several moments later, the switch is opened. Current is induced in the loop by the solenoid

1. only when the switch is first closed.
2. only when the switch is opened.
3. only when the switch is closed and opened.
4. during the time the switch is closed.

Reading Question 20-1 (2 of 2)

A loop is placed next to a circuit containing a solenoid and a switch. The circuit is attached to an AC source and then the switch is closed. Several moments later, the switch is opened. Current is induced in the loop by the solenoid

1. only when the switch is first closed.
2. only when the switch is opened.
3. only when the switch is closed and opened.
4. **Answer: during the time the switch is closed.**

Reading Question 20-2 (1 of 2)

The flux of the magnetic field caused by an induced current in a loop

1. is in the same direction as the flux causing the induced current.
2. is in the opposite direction to the flux causing the induced current.
3. is in the same direction as the change in flux causing the induced current.
4. is in the opposite direction to the change in flux causing the induced current.

Reading Question 20-2 (2 of 2)

The flux of the magnetic field caused by an induced current in a loop

1. is in the same direction as the flux causing the induced current.
2. is in the opposite direction to the flux causing the induced current.
3. is in the same direction as the change in flux causing the induced current.
4. **Answer: is in the opposite direction to the change in flux causing the induced current.**

Reading Question 20-3 (1 of 2)

A loop of wire lies on the table. The south end of a magnet is moved toward the loop from above. As viewed from above, in which directions are the induced current and the induced magnetic field?

1. counter-clockwise, up
2. counter-clockwise, down
3. clockwise, up
4. clockwise, down

Reading Question 20-3 (2 of 2)

A loop of wire lies on the table. The south end of a magnet is moved toward the loop from above. As viewed from above, in which directions are the induced current and the induced magnetic field?

1. counter-clockwise, up
2. counter-clockwise, down
3. clockwise, up
4. **Answer: clockwise, down**

Reading Question 20-4 (1 of 2)

A magnet mounted on a pendulum swings back and forth. The induced current at the nearby wire loop is at its maximum when placed

1. at the top of the magnet's swing on the right.
2. at the top of the magnet's swing on the left.
3. at the bottom of the magnet's swing.
4. where the magnet is in between the top of its swing and the bottom.

Reading Question 20-4 (2 of 2)

A magnet mounted on a pendulum swings back and forth. The induced current at the nearby wire loop is at its maximum when placed

1. at the top of the magnet's swing on the right.
2. at the top of the magnet's swing on the left.
3. **Answer: at the bottom of the magnet's swing.**
4. where the magnet is in between the top of its swing and the bottom.

Reading Question 20-5 (1 of 2)

In which of the following cases is the back emf of a motor the greatest?

1. when turned off
2. when just starting up
3. when running under a large mechanical load
4. when running under no mechanical load

Reading Question 20-5 (2 of 2)

In which of the following cases is the back emf of a motor the greatest?

1. when turned off
2. when just starting up
3. when running under a large mechanical load
4. **Answer: when running under no mechanical load**

Reading Question 20-6 (1 of 2)

At $t = 0$, the switch in an RL circuit is closed. At that moment,

1. $I = 0, \varepsilon_L = -\varepsilon$
2. $I = 0, \varepsilon_L = 0$
3. $I = \varepsilon / R, \varepsilon_L = \varepsilon$
4. $I = \varepsilon / R, \varepsilon_L = 0$

Reading Question 20-6 (2 of 2)

At $t = 0$, the switch in an RL circuit is closed. At that moment,

1. **Answer:** $I = 0, \varepsilon_L = -\varepsilon$

2. $I = 0, \varepsilon_L = 0$

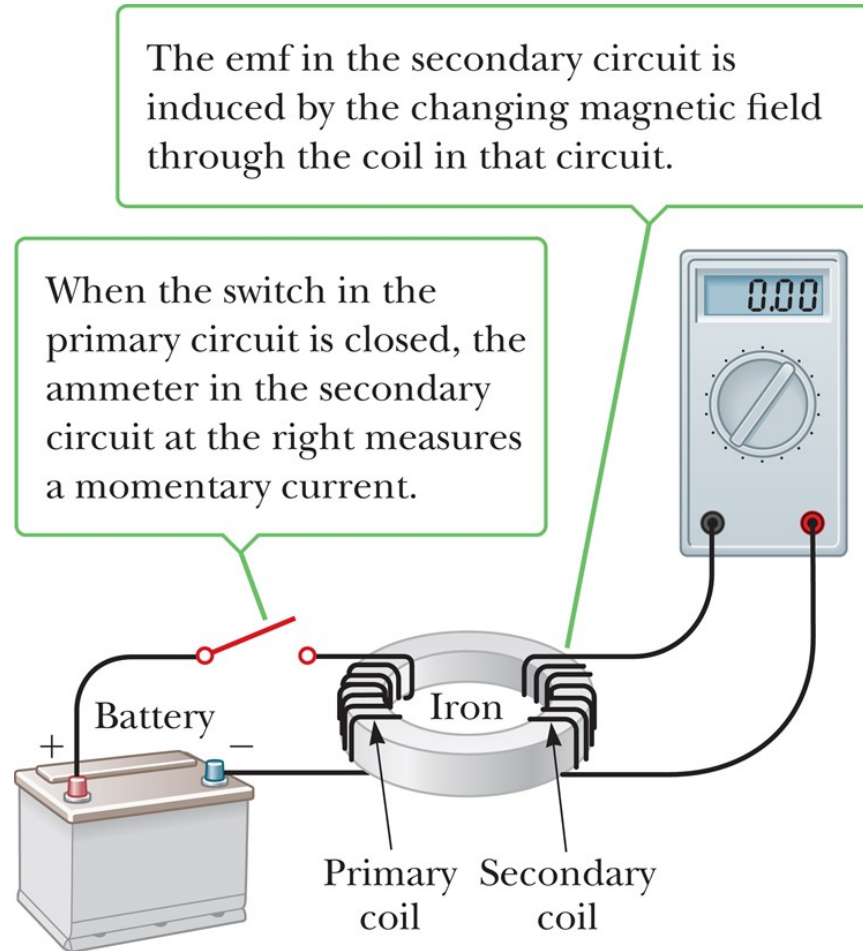
3. $I = \varepsilon / R, \varepsilon_L = \varepsilon$

4. $I = \varepsilon / R, \varepsilon_L = 0$

Topic 20: Induced Voltages and Inductance

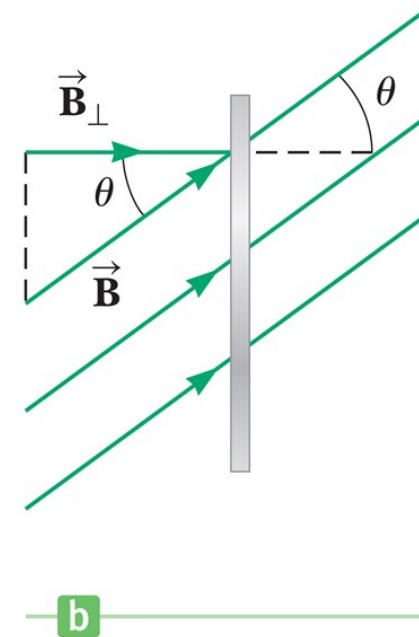
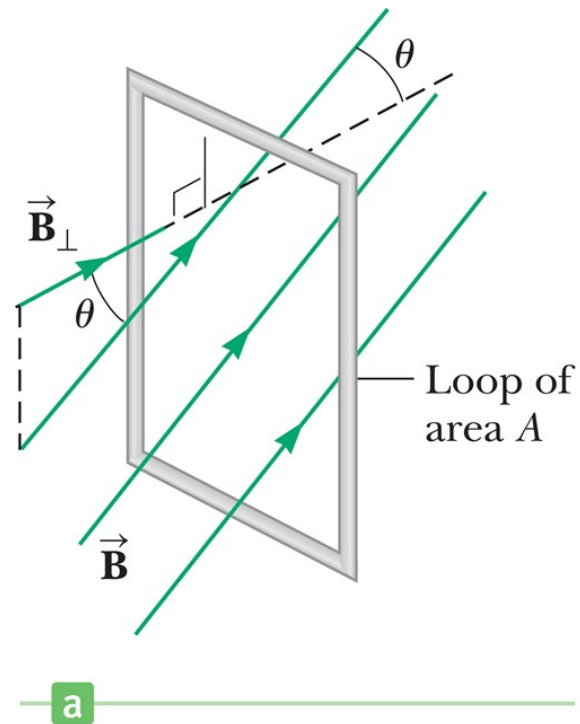
TOPIC DISCUSSION

Induced emf and Magnetic Flux

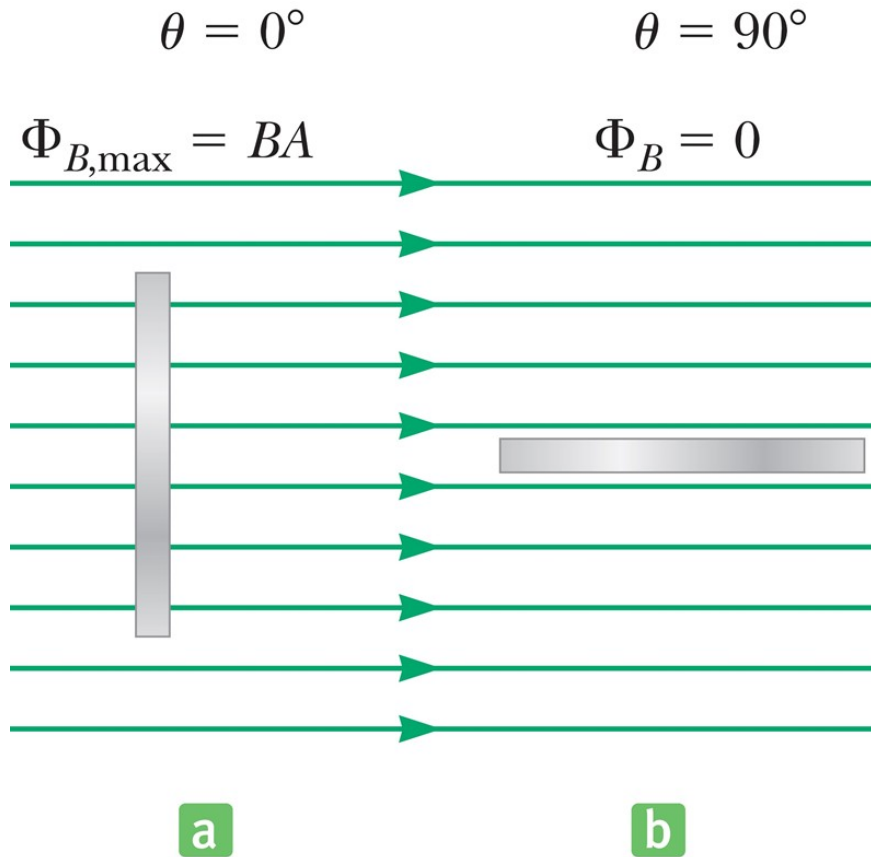


Magnetic Flux (1 of 2)

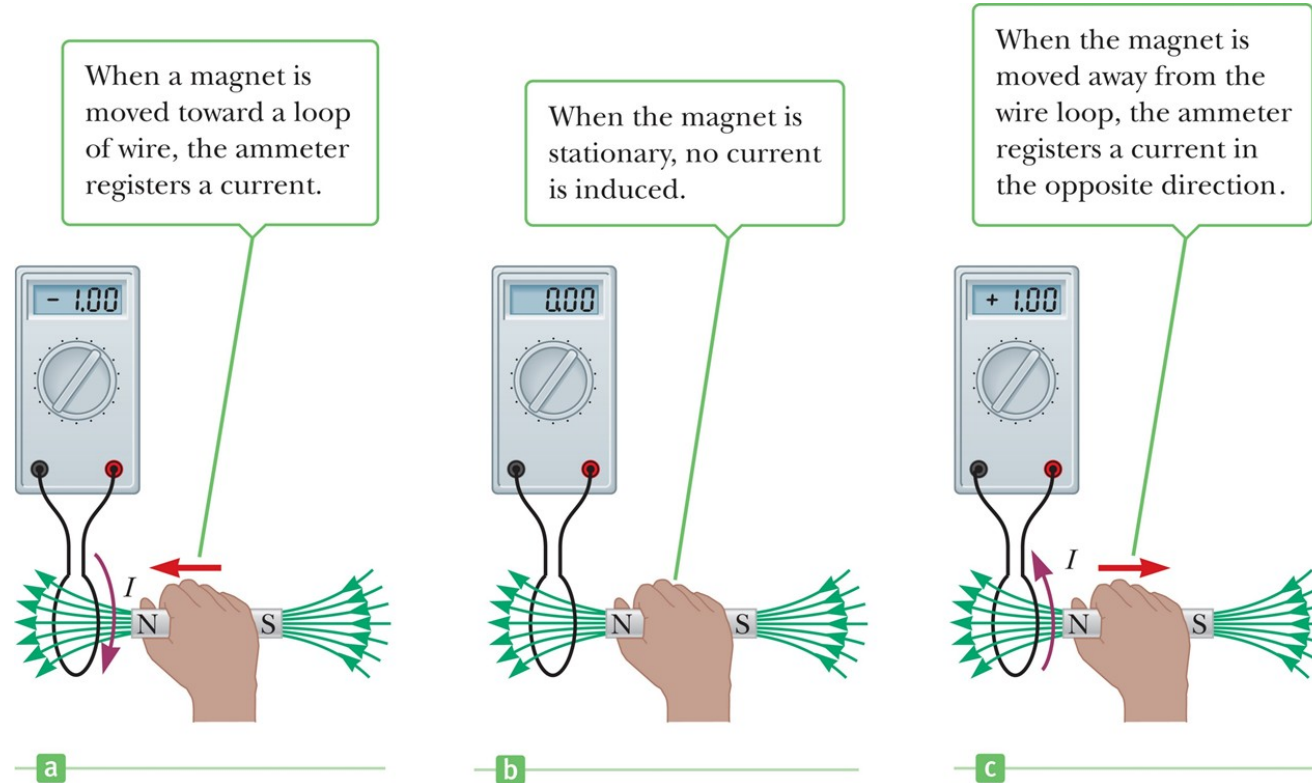
$$\Phi_B \equiv B_{\perp} A = BA \cos \theta \quad \text{SI unit: T} \cdot \text{m}^2 \text{ or Wb}$$



Magnetic Flux (2 of 2)



Faraday's Law of Induction and Lenz's Law (1 of 3)



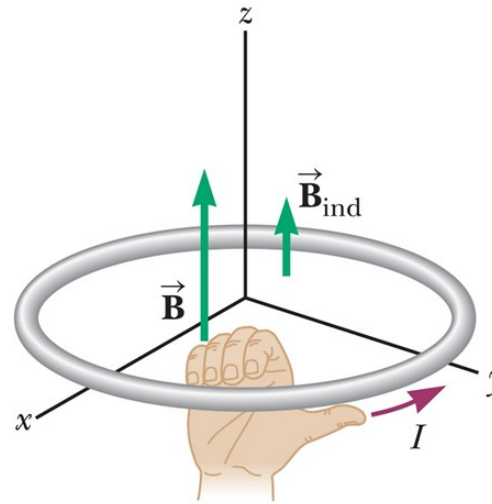
Faraday's Law of Induction and Lenz's Law (2 of 3)

$$\mathcal{E} = -N \frac{\Delta \Phi_B}{\Delta t}$$

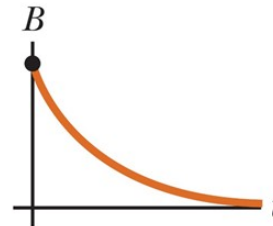
$$\Phi_B = BA \cos \theta$$

Lenz's law: The current caused by the induced emf travels in the direction that creates a magnetic field with flux opposing the change in the original flux through the circuit.

Faraday's Law of Induction and Lenz's Law (3 of 3)



a

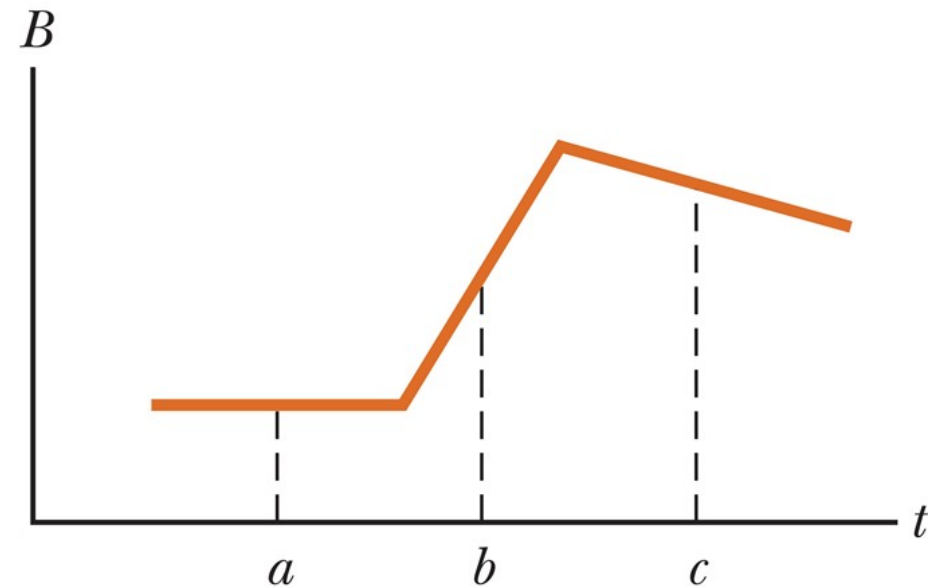


b

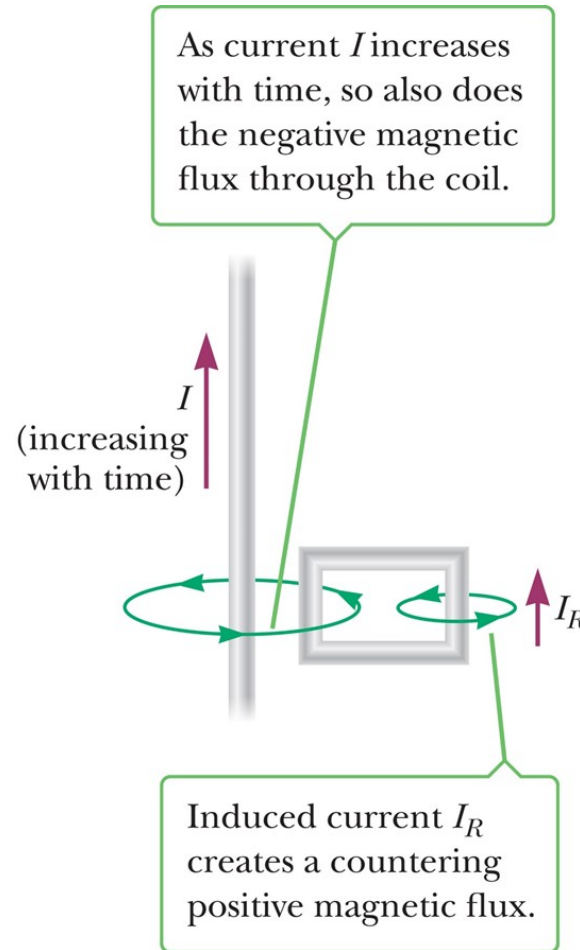
Think – Pair – Share 1

The figure below is a graph of the magnitude B versus time for a magnetic field that passes through a fixed loop and is oriented perpendicular to the plane of the loop. Rank the magnitudes of the emf generated in the loop at the three instants indicated from largest to smallest.

1. $c > b > a$
2. $a > b > c$
3. $b > c > a$
4. $b > a > c$

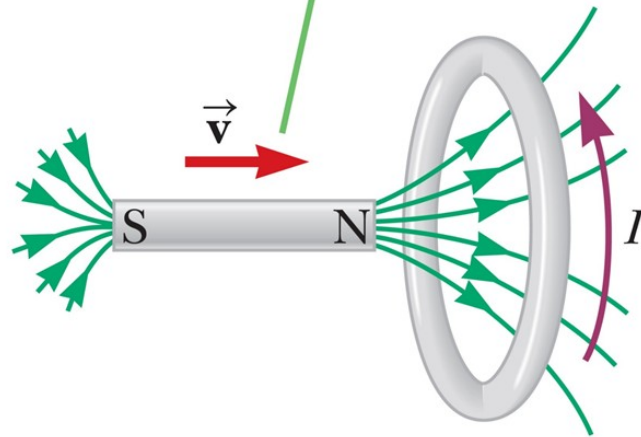


Finding the Direction of the Induced Current (1 of 7)



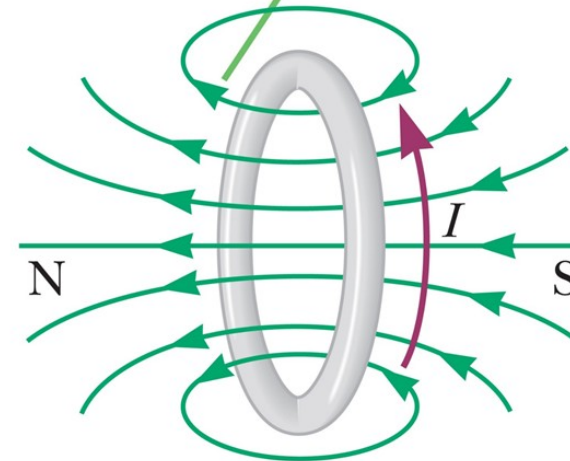
Finding the Direction of the Induced Current (2 of 7)

As the magnet moves to the right, positive flux increases through the coil.



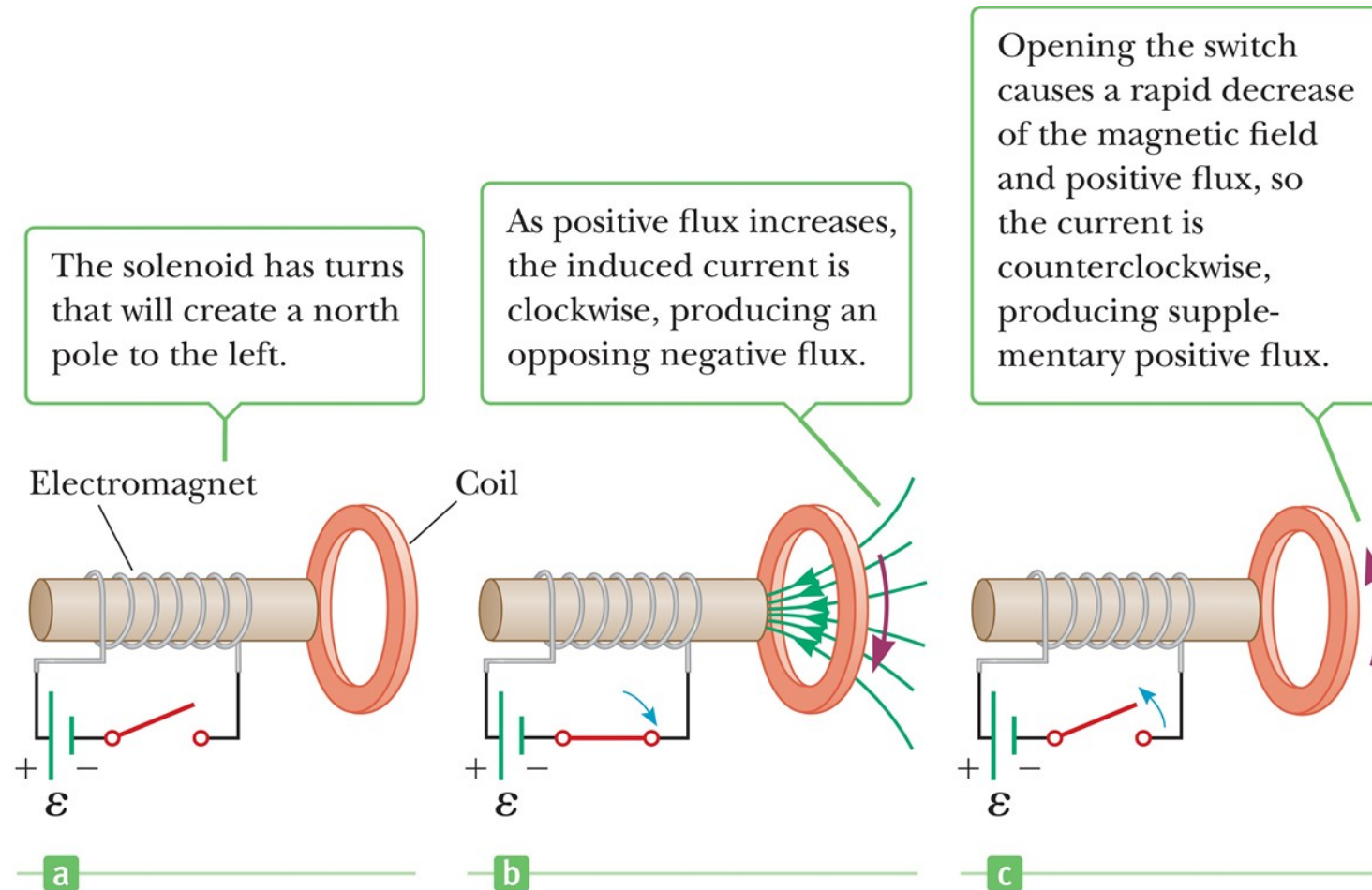
a

The induced current produces negative flux, countering the increasing positive flux.



b

Finding the Direction of the Induced Current (3 of 7)



Think – Pair – Share 2

A bar magnet is falling toward the center of a loop of wire, with the north pole oriented downward. Viewed from the same side of the loop as the magnet, as the north pole approaches the loop, what is the direction of the induced current?

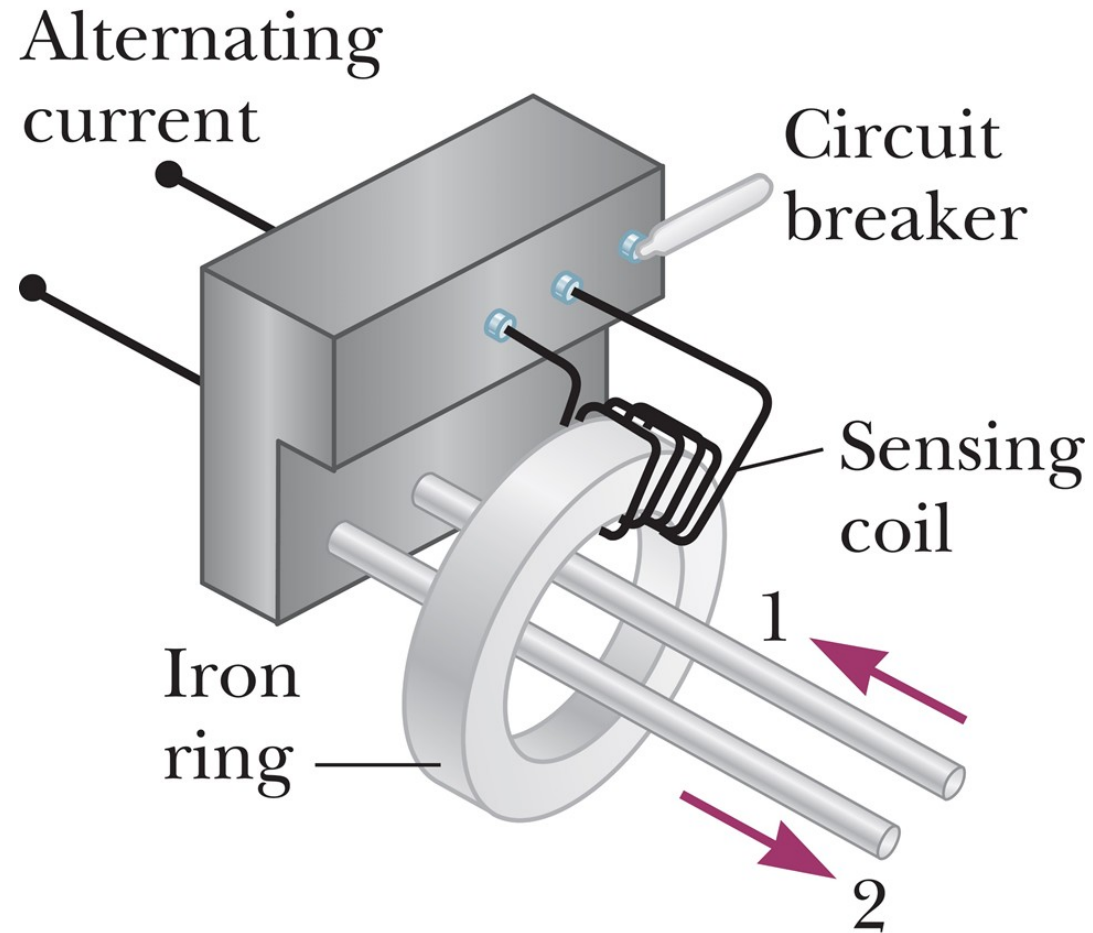
1. clockwise
2. zero
3. counterclockwise
4. along the length of the magnet

Think – Pair – Share 3

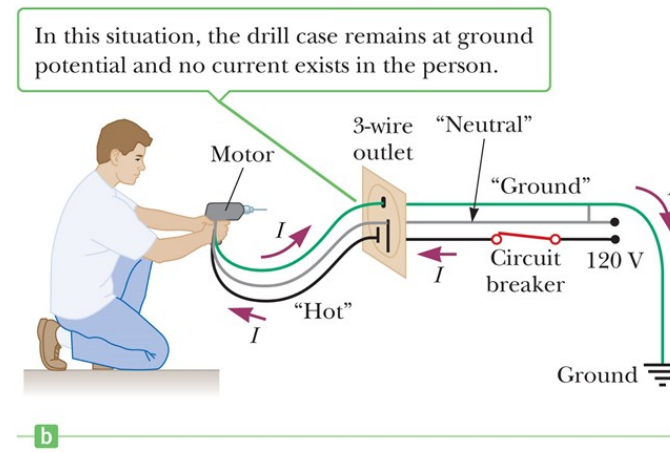
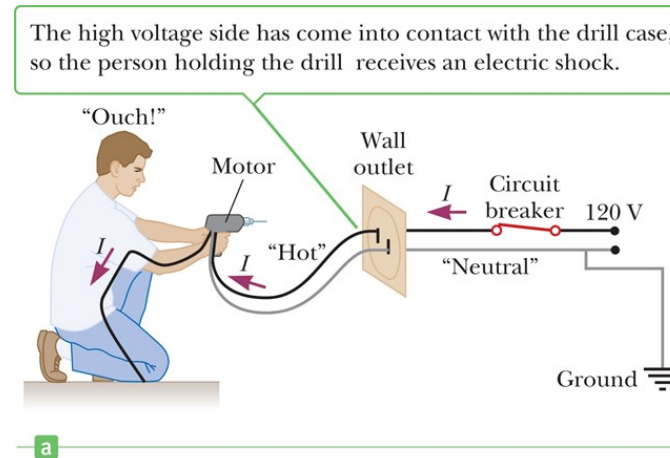
Two circular loops are side by side and lie in the xy -plane. A switch is closed, starting a counterclockwise current in the left-hand loop, as viewed from a point on the positive z -axis passing through the center of the loop. Which of the following statements is true of the right-hand loop?

1. The current remains zero.
2. An induced current moves counterclockwise.
3. An induced current moves clockwise.

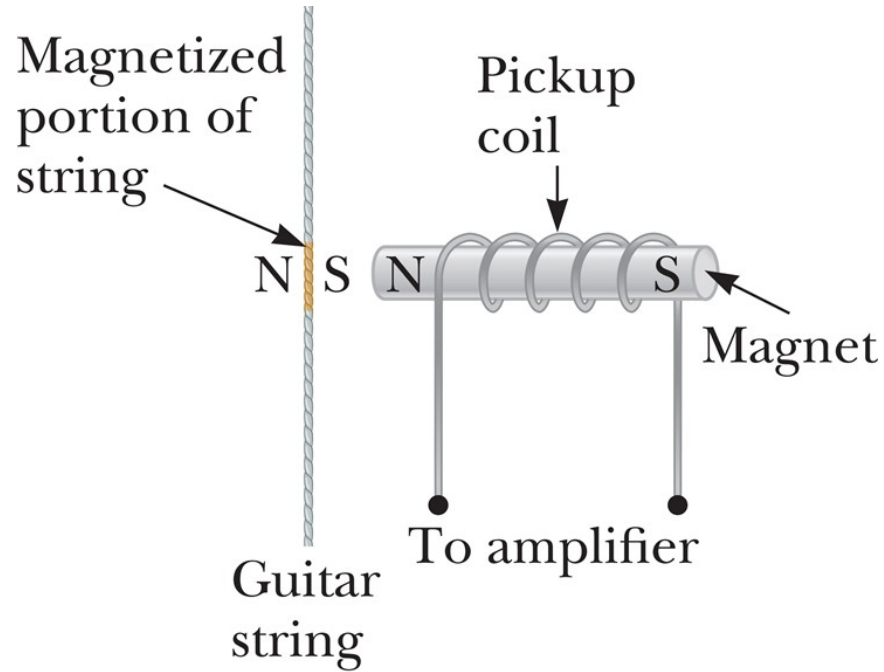
Finding the Direction of the Induced Current (4 of 7)



Finding the Direction of the Induced Current (5 of 7)



Finding the Direction of the Induced Current (6 of 7)



a



b

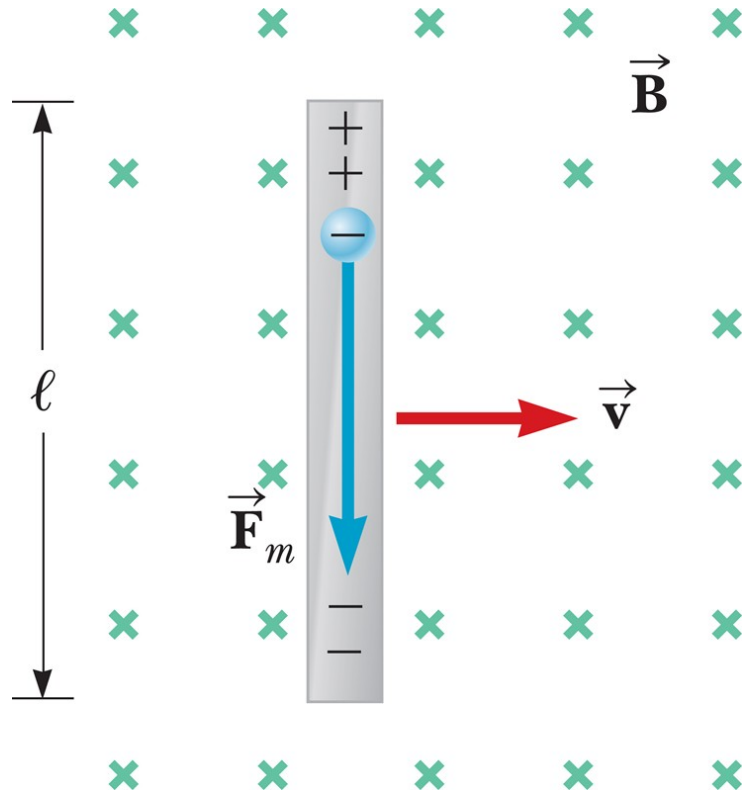
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Finding the Direction of the Induced Current (7 of 7)



Courtesy of PedsLink Pediatric Healthcare Resources,
Newport Beach, CA

Motional emf (1 of 3)

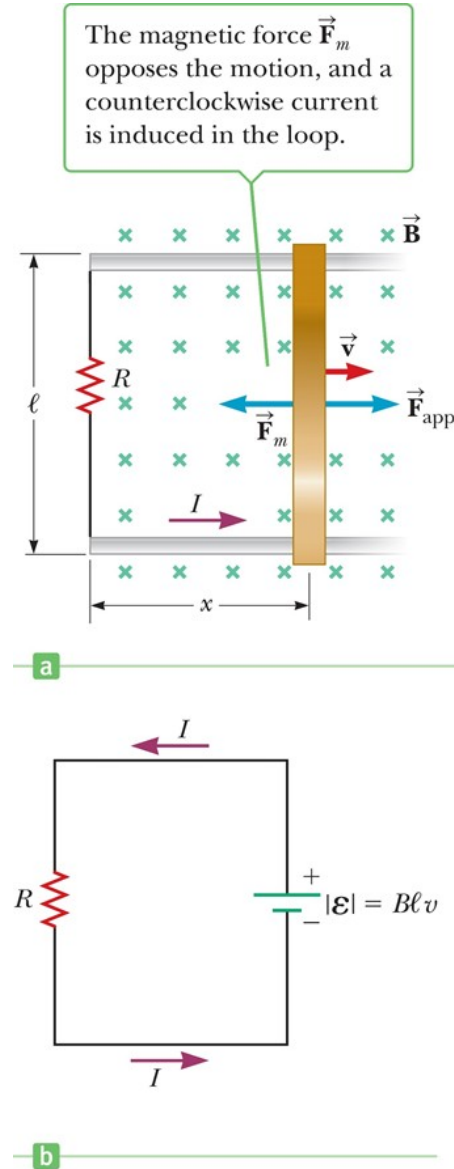


$$F_m = qvB$$

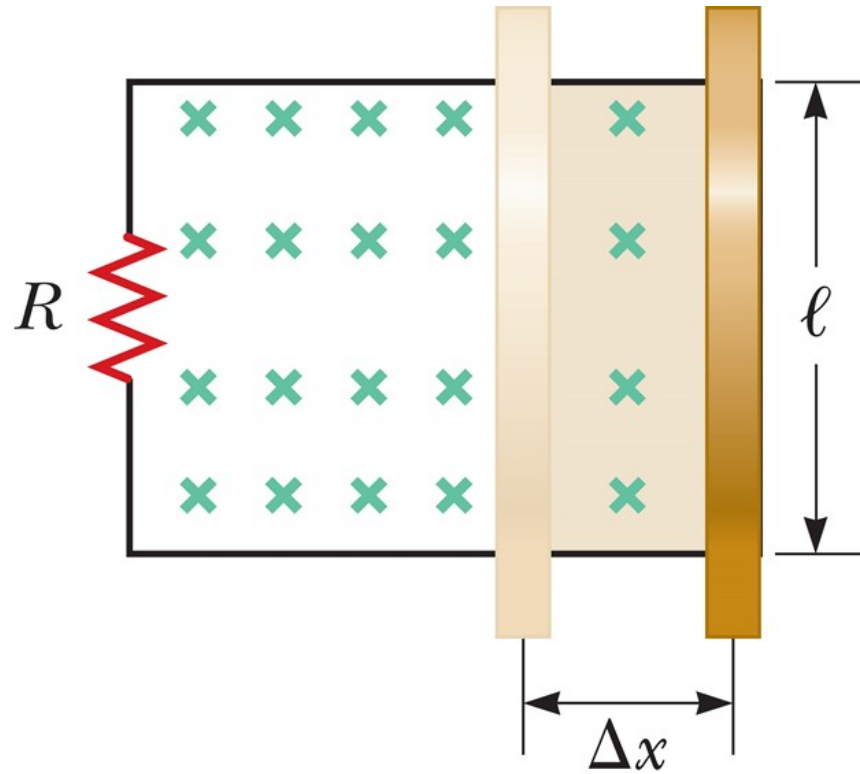
$$qE = qvB \quad \text{or} \quad E = vB$$

$$\Delta V = E\ell = B\ell v$$

Motional emf (2 of 3)



Motional emf (3 of 3)



$$\Delta\Phi_B = BA = B\ell\Delta x$$

$$|\mathcal{E}| = \frac{\Delta\Phi_B}{\Delta t} = B\ell \frac{\Delta x}{\Delta t} = B\ell v$$

$$I = \frac{|\mathcal{E}|}{R} = \frac{B\ell v}{R}$$

Think – Pair – Share 4

A horizontal metal bar oriented east-west drops straight down in a location where Earth's magnetic field is due north. As a result, an emf develops between the ends. Which end is positively charged?

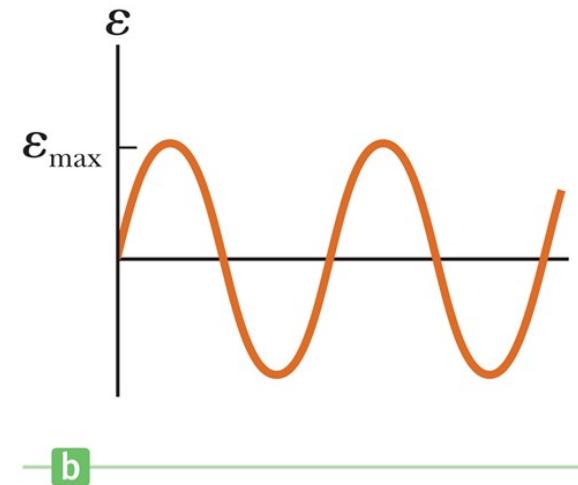
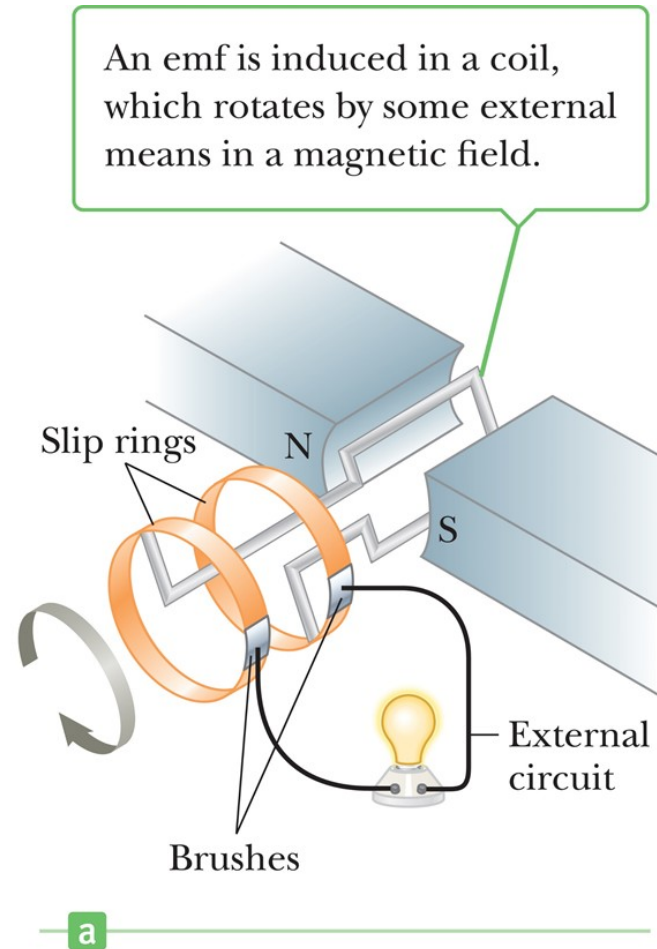
1. the east end
2. the west end
3. neither end carries a current

Think – Pair – Share 5

You intend to move a rectangular loop of wire into a region of uniform magnetic field at a given speed so as to induce an emf in the loop. The plane of the loop must remain perpendicular to the magnetic field lines. In which orientation should you hold the loop while you move it into the region with the magnetic field to generate the largest emf?

1. with the long dimension of the loop parallel to the velocity vector
2. with the short dimension of the loop parallel to the velocity vector
3. Either way because the emf is the same regardless of orientation

Generators (1 of 4)

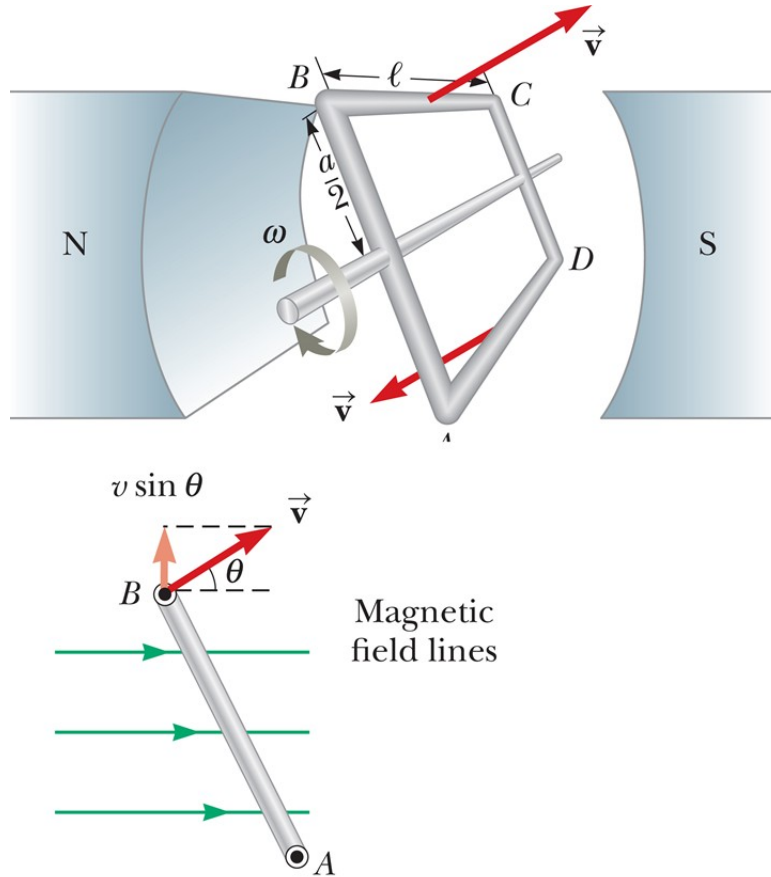


Generators (2 of 4)



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Generators (3 of 4)



$$\mathcal{E} = B\ell v$$

$$\mathcal{E} = 2B\ell v_{\perp} = 2B\ell v \sin \theta$$

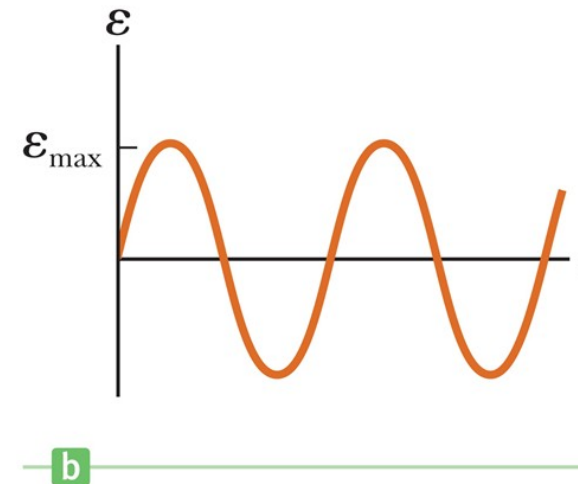
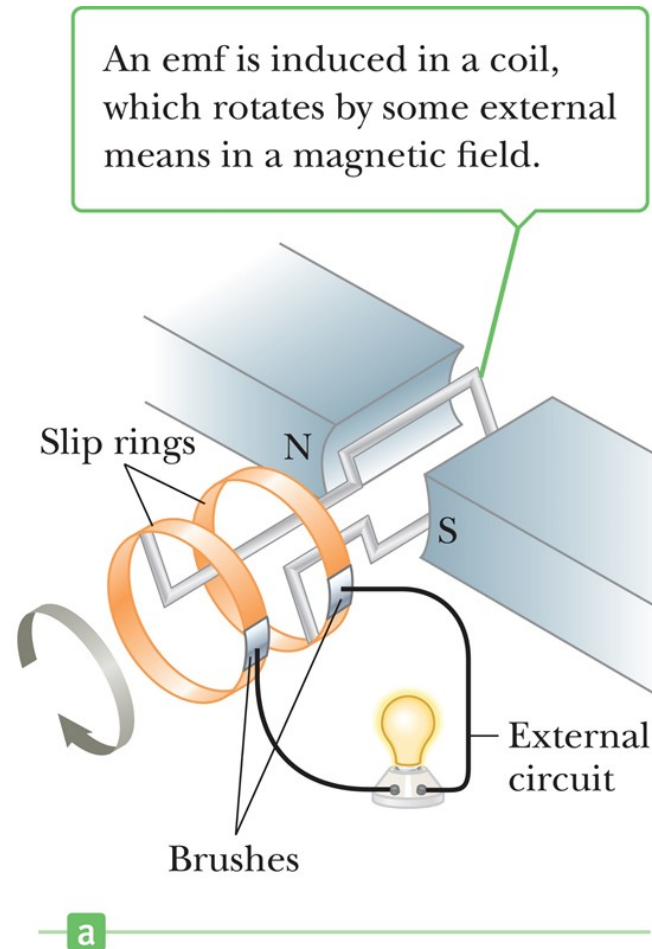
$$v = r\omega = \left(\frac{a}{2}\right)\omega$$

$$\mathcal{E} = 2B\ell \left(\frac{a}{2}\right)\omega \sin \omega t = B\ell a \omega \sin \theta$$

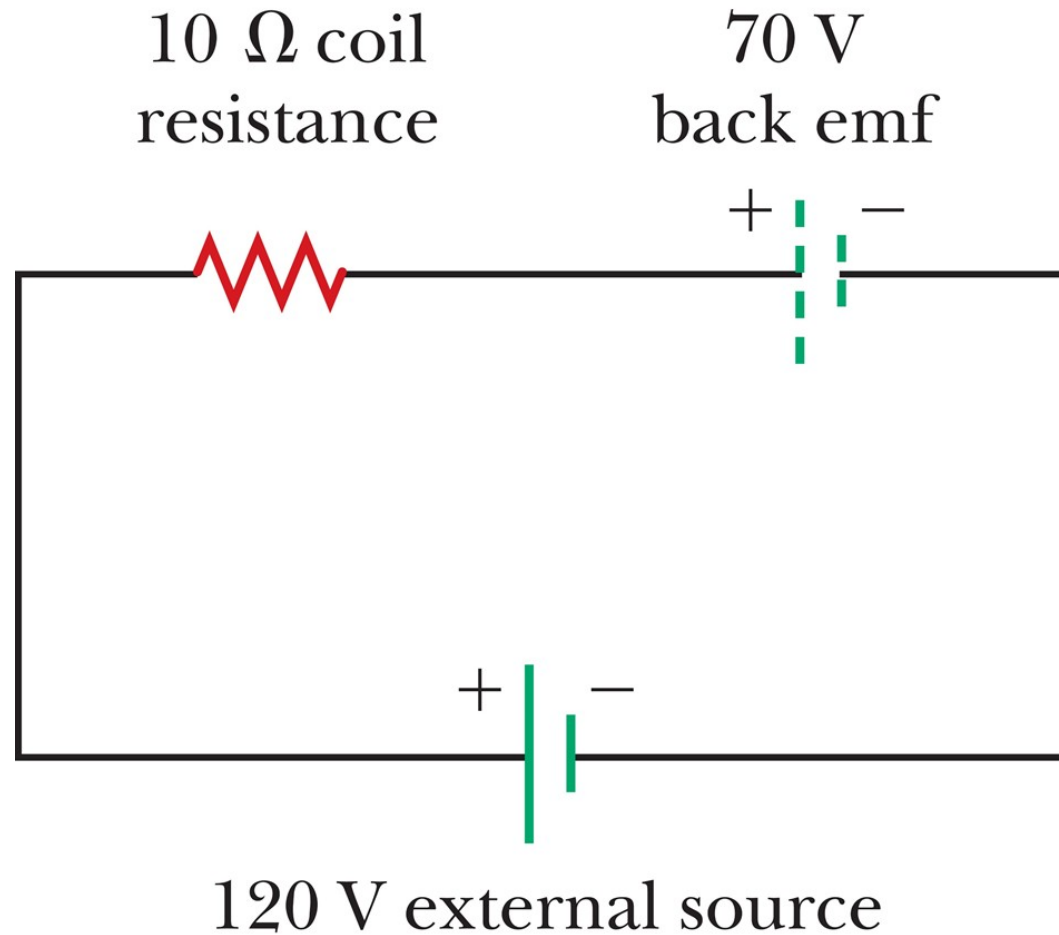
$$\mathcal{E} = NBA \omega \sin \omega t$$

$$\mathcal{E}_{\max} = NBA\omega$$

Generators (4 of 4)

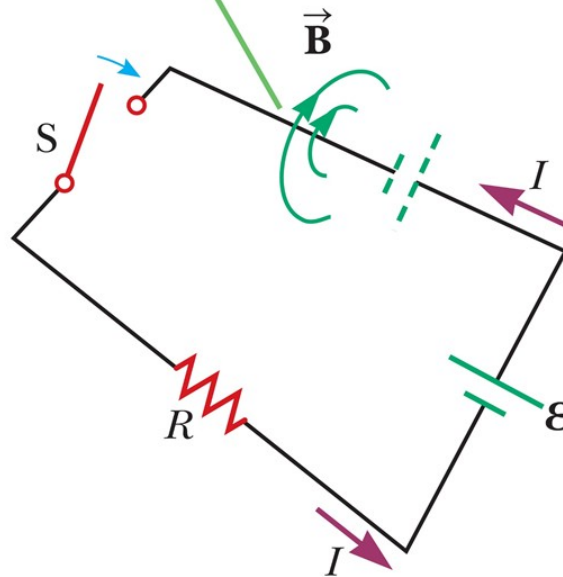


Motors and Back emf

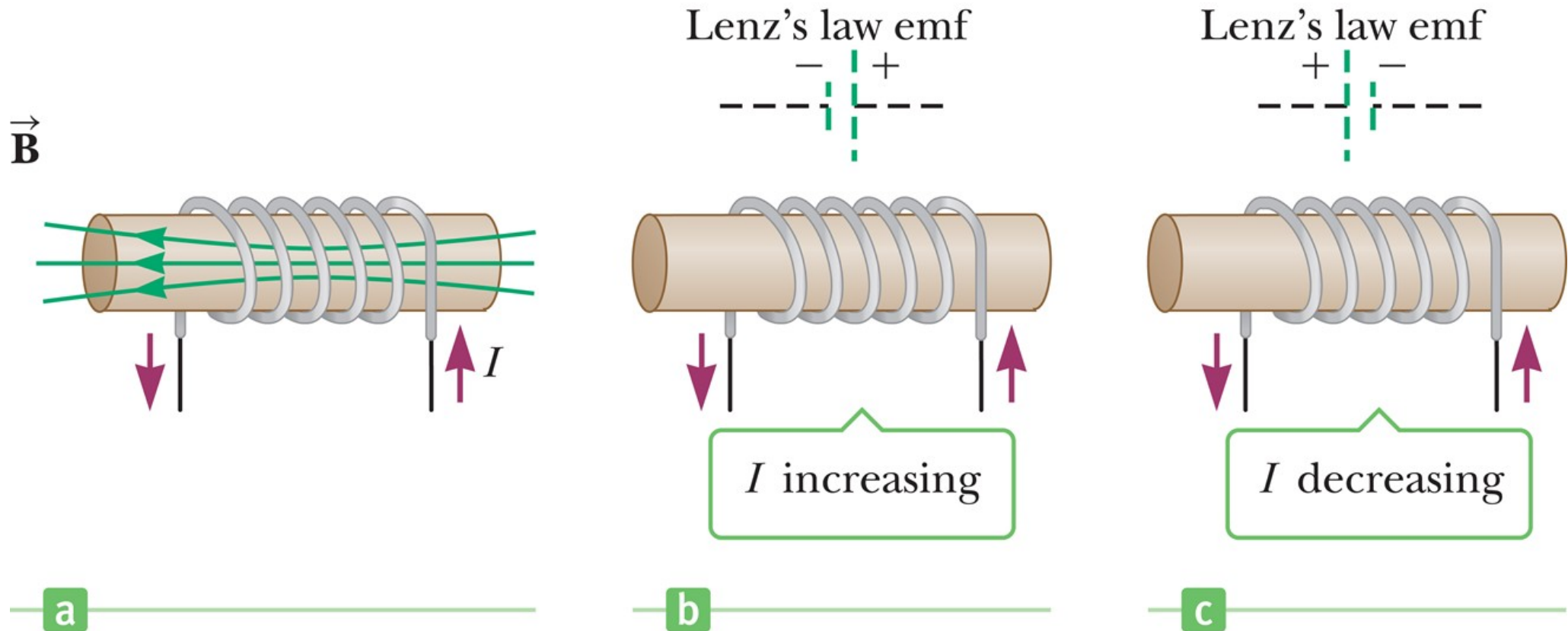


Self-Inductance (1 of 3)

As the current increases toward its maximum value it creates changing magnetic flux, inducing an opposing emf in the loop.



Self-Inductance (2 of 3)



Self-Inductance (3 of 3)

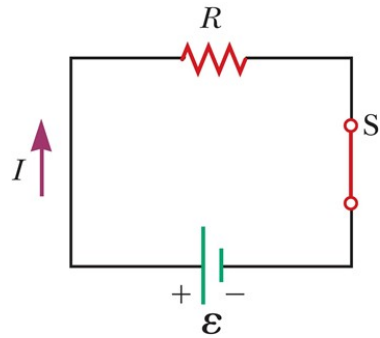
$$\mathcal{E} = -N \frac{\Delta \Phi_B}{\Delta t} \qquad \mathcal{E} \equiv -L \frac{\Delta I}{\Delta t}$$

$$1 \text{ H} = 1 \text{ V} \cdot \text{s} / \text{A}$$

$$N \frac{\Delta \Phi_B}{\Delta t} = L \frac{\Delta I}{\Delta t}$$

$$L = N \frac{\Delta \Phi_B}{\Delta I} = \frac{N \Phi_B}{I}$$

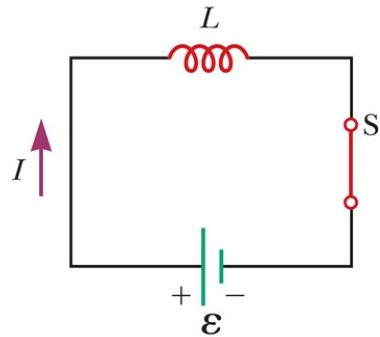
RL Circuits (1 of 2)



a

$$\mathcal{E} - IR = 0$$

$$\Delta V_R = -IR$$

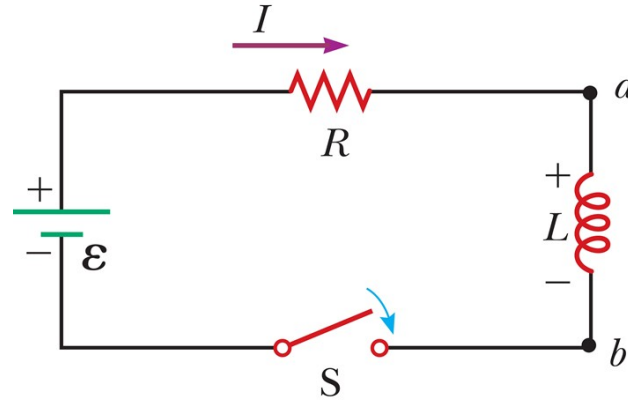
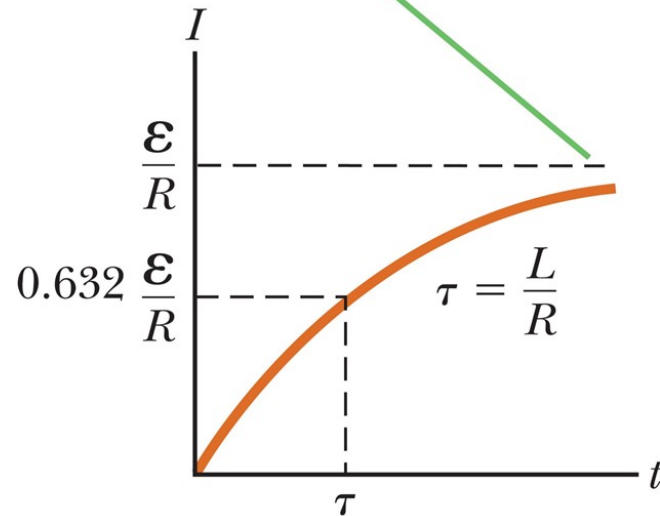


b

$$\mathcal{E}_L = -L \frac{\Delta I}{\Delta t}$$

RL Circuits (2 of 2)

After the switch is closed at $t = 0$, the current increases toward its maximum value $\mathcal{E}/R$.



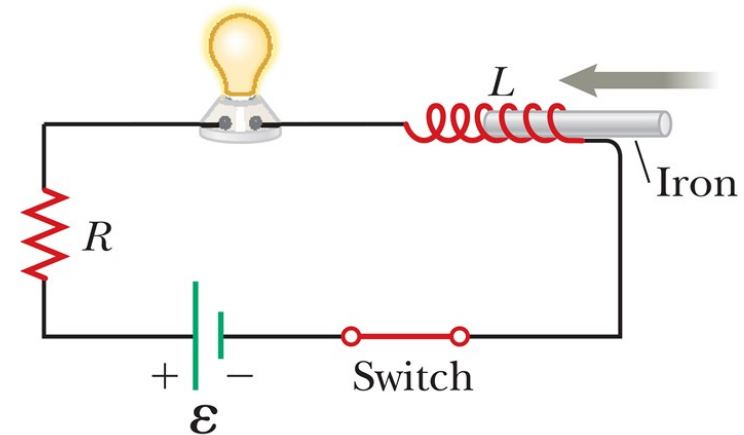
$$\mathcal{E}_L = -L \frac{\Delta I}{\Delta t} \quad \tau = \frac{L}{R}$$

$$I = \frac{\mathcal{E}}{R} \left(1 - e^{-t/\tau} \right)$$

Think – Pair – Share 6

The switch in the circuit shown in the figure below is closed, and the lightbulb glows steadily. The inductor is a simple air-core solenoid. An iron rod is inserted into the interior of the solenoid, increasing the magnitude of the magnetic field in the solenoid. As the rod is inserted, the brightness of the lightbulb

1. increases.
2. decreases.
3. remains the same.



Energy Stored in Magnetic Fields

$$PE_L = \frac{1}{2}LI^2$$

$$PE_C = \frac{1}{2}C(\Delta V)^2$$